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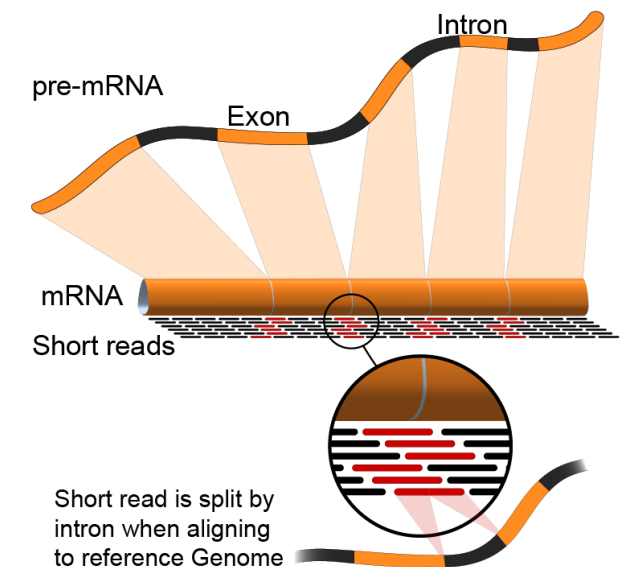
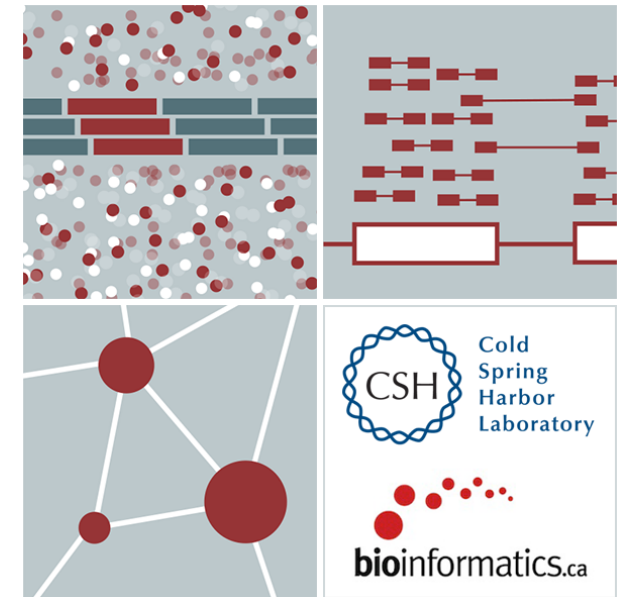
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Module 1: Introduction to RNA sequencing

Malachi Griffith & Obi Griffith

RNA-seq Analysis
June 22-24, 2026



Learning objectives of the course



- **Module 1: Introduction to RNA Sequencing**
- Module 2: Alignment and Visualization
- Module 3: Expression and Differential Expression
- Module 4: Alignment Free Expression Estimation
- Tutorials
 - Provide a working example of an RNA-seq analysis pipeline
 - Run in a ‘reasonable’ amount of time with modest computer resources
 - Self contained, self explanatory, portable



Learning objectives of module 1

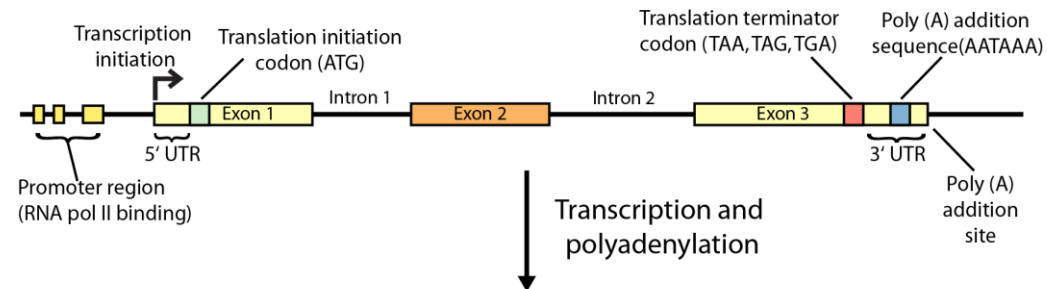
- Introduction to the theory and practice of RNA sequencing (RNA-seq) analysis
 - Background molecular biology
 - Challenges specific to RNA-seq
 - General goals and themes of RNA-seq analysis workflows
 - Common technical questions related to RNA-seq analysis
 - Introduction to the RNA-seq hands on tutorial

Gene expression

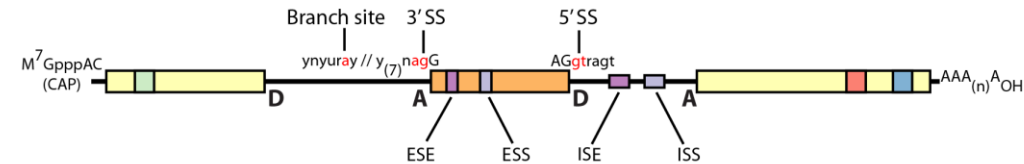
Thinking about the molecular biology here, what is actually being sequenced in an RNA-seq experiment?

Does it differ depending on the sequencing platform? Or for bulk vs single cell sequencing?

Double-stranded genomic DNA template

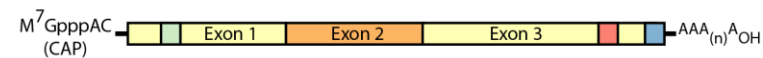


Single-stranded pre-mRNA (nuclear RNA)



RNA processing

Mature mRNA

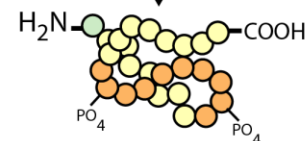


Export to cytoplasm and translation

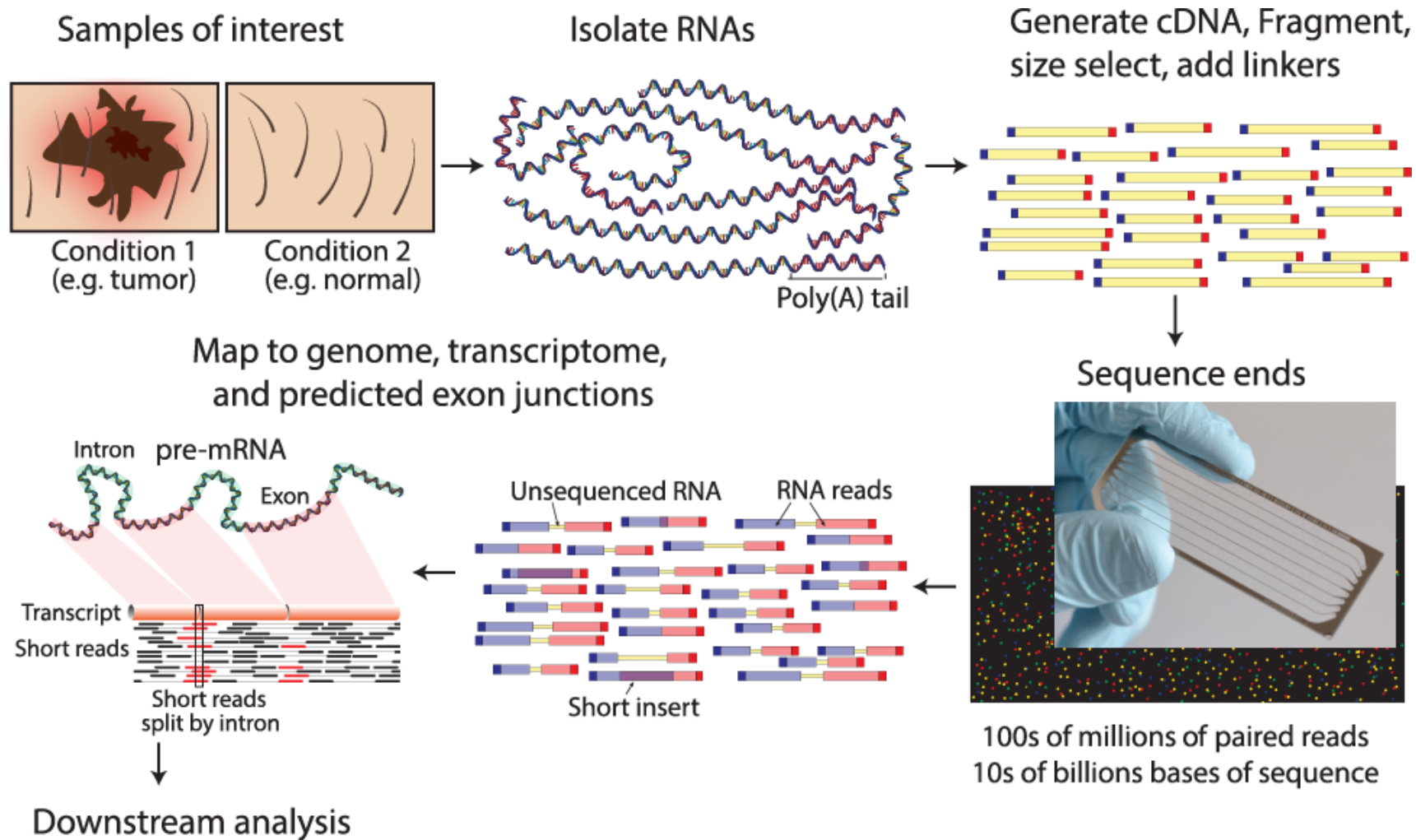
Protein (amino acid sequence)



Folding, posttranslational modification, subcellular localization, etc.



RNA sequencing





MPS (NGS) Platforms: Illumina is currently dominant

Production-scale sequencers

Key specifications				
	NextSeq 1000 and 2000 Systems	NovaSeq 6000 System	NovaSeq X Series	
	Max output per flow cell	540 Gb ^a	3 Tb ^b	8 Tb ^c
	Run time (range) ^d	~8–44 hr	~13–44 hr	~17–48 hr
	Max reads per run (single reads)	1.8B ^a	10B (single flow cell) ^b 20B (dual flow cells)	26B (single flow cell) ^c 52B (dual flow cells) ^{c,e}
Max read length	2 × 300 bp	2 × 250 bp	2 × 150 bp	

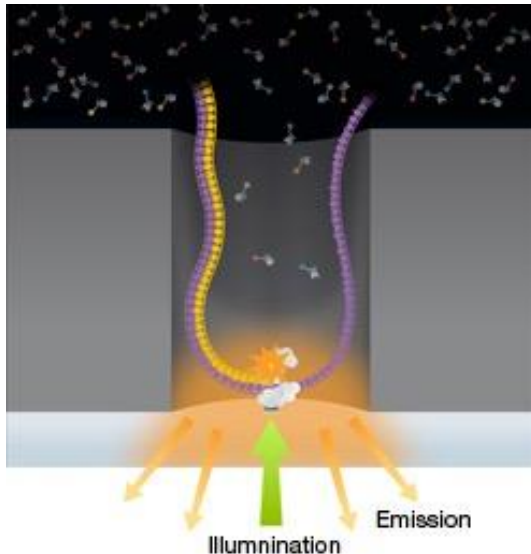
- Higher accuracy, range of capacity and throughput
- Slightly longer read lengths on some platforms

Next-next (3rd) generation sequencing platforms

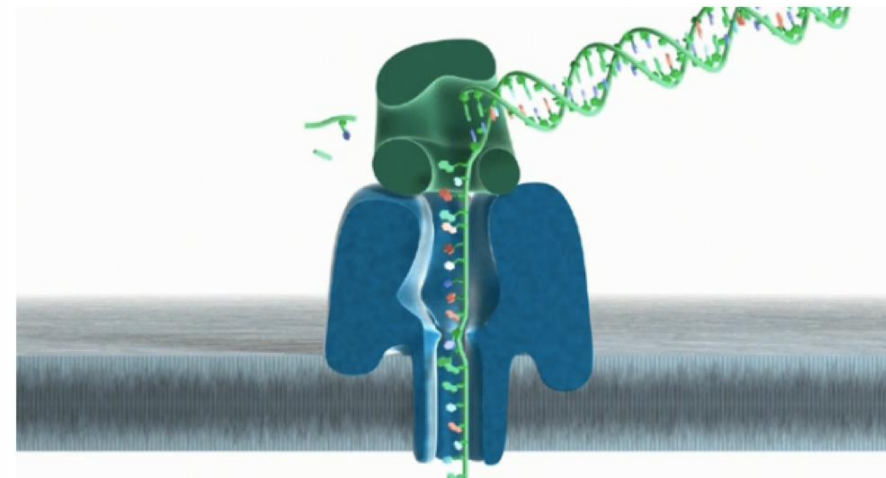


Defining Characteristics: Long reads (10-100 kb) from single molecules.

Pacific Biosciences: watching a polymerase synthesize DNA/cDNA in real time



Oxford Nanopore: Translocating DNA/RNA through a nanopore with electrode-based detection



The promise: Long reads will allow us to accurately sequence and assemble whole human genomes, from scratch, without using the reference genome.

Status: Currently limited by lower throughput, higher base error rate and higher cost. 3rd generation technologies have proven useful, but generally for niche applications so far.

Challenges

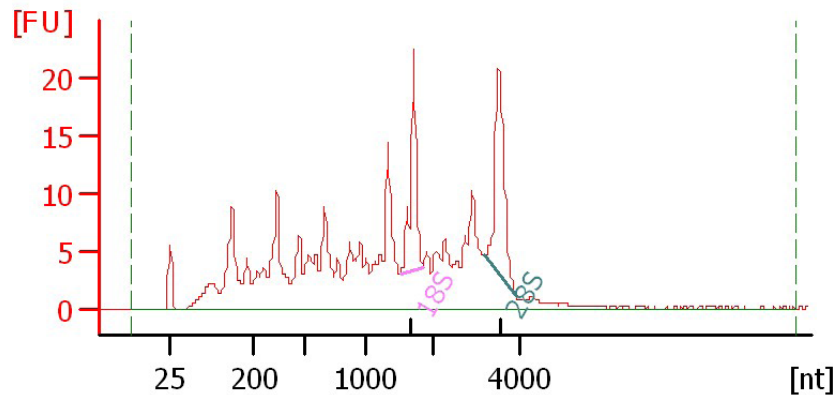


- Sample
 - Purity?, quantity?, quality?
- RNAs consist of small exons that may be separated by large introns
 - Mapping reads to genome is challenging
- The relative abundance of RNAs vary wildly
 - $10^5 - 10^7$ orders of magnitude
 - Since RNA sequencing works by random sampling, a small fraction of highly expressed genes may consume the majority of reads
 - Ribosomal and mitochondrial genes
- RNAs come in a wide range of sizes
 - Small RNAs must be captured separately
 - PolyA selection of large RNAs may result in 3' end bias
- RNA is fragile compared to DNA (easily degraded)

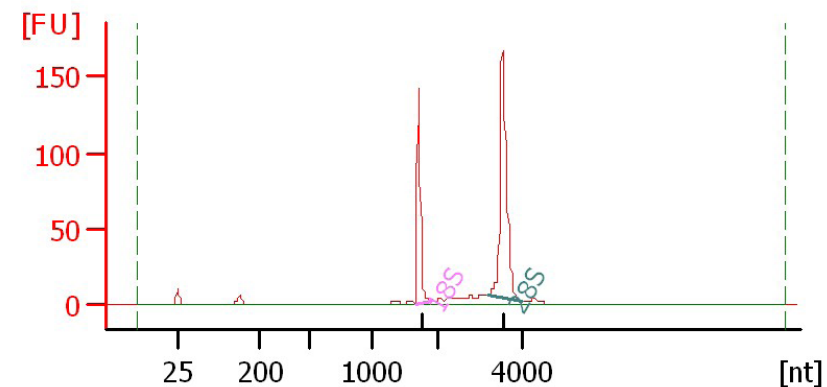
Agilent example / interpretation



- <https://goo.gl/uC5a3C>
- 'RIN' = RNA integrity number
 - 0 (bad) to 10 (good)



RIN = 6.0



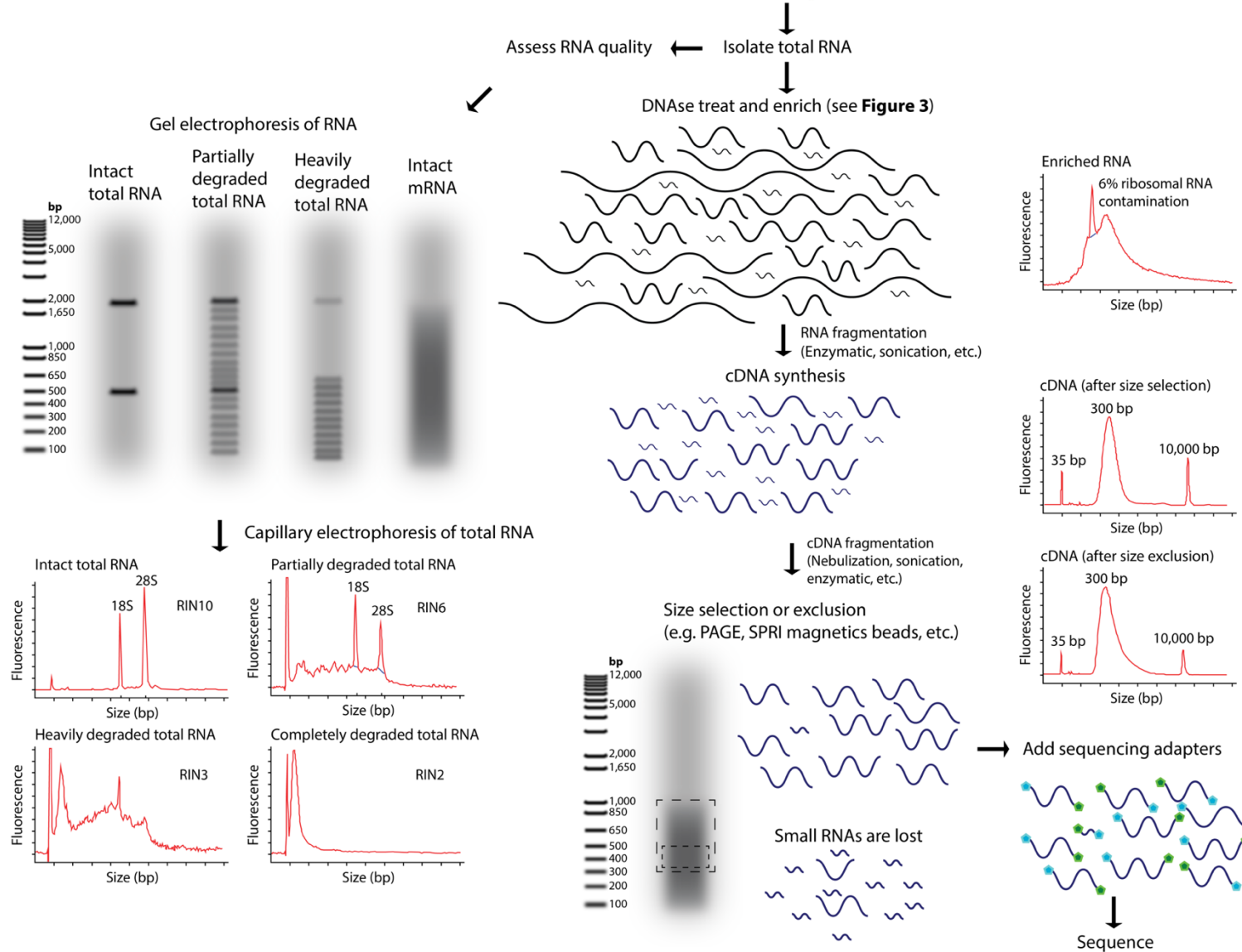
RIN = 10



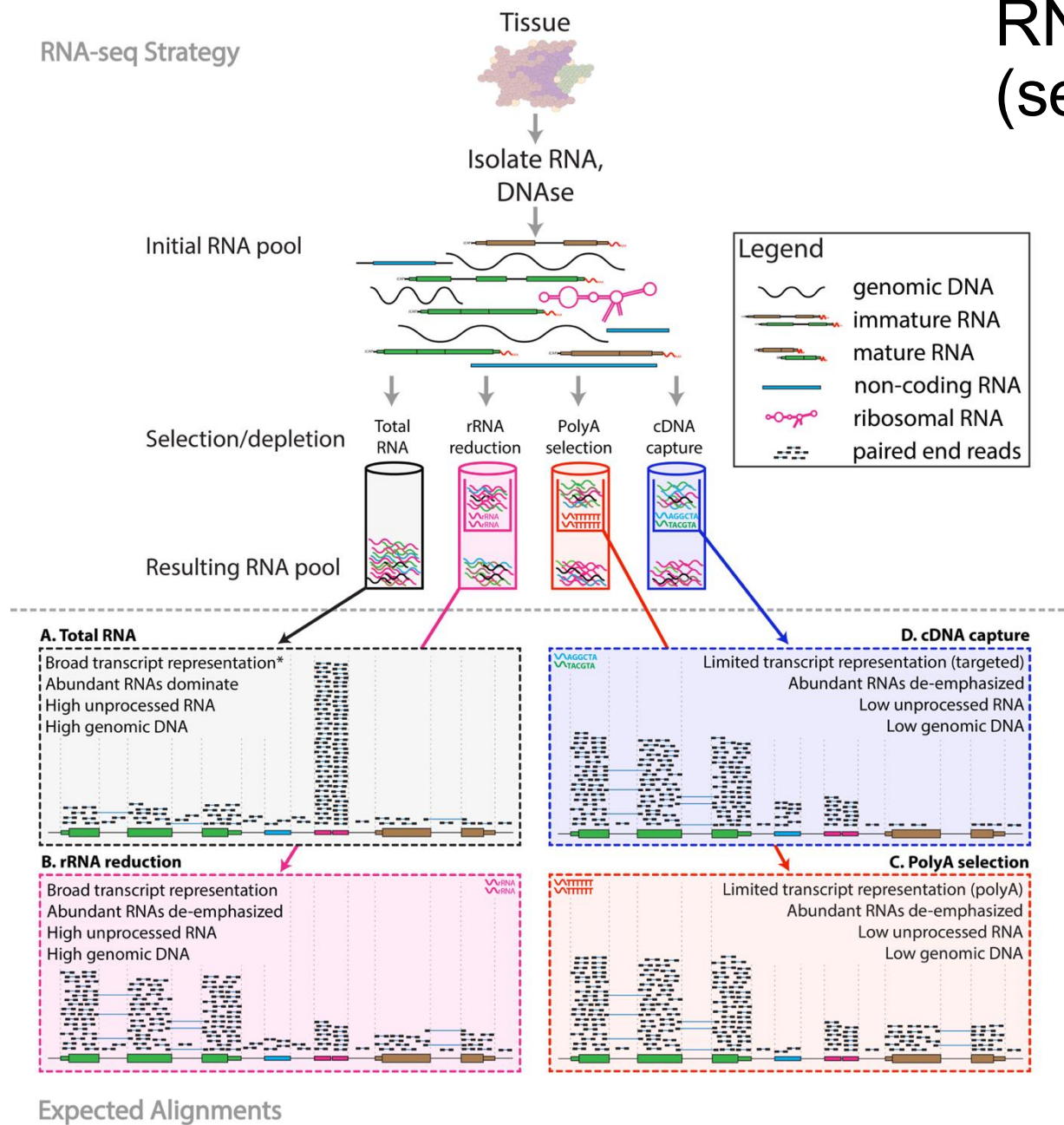
There are many RNA-seq library construction strategies

- Total RNA versus polyA+ RNA?
- Ribo-reduction?
- Size selection (before and/or after cDNA synthesis)
 - Small RNAs (microRNAs) vs. large RNAs?
 - A narrow fragment size distribution vs. a broad one?
- Linear amplification?
- Stranded vs. un-stranded libraries
- Exome captured vs. un-captured
- Library normalization?
- These details can affect analysis strategy
 - Especially comparisons between libraries

Fragmentation and size selection



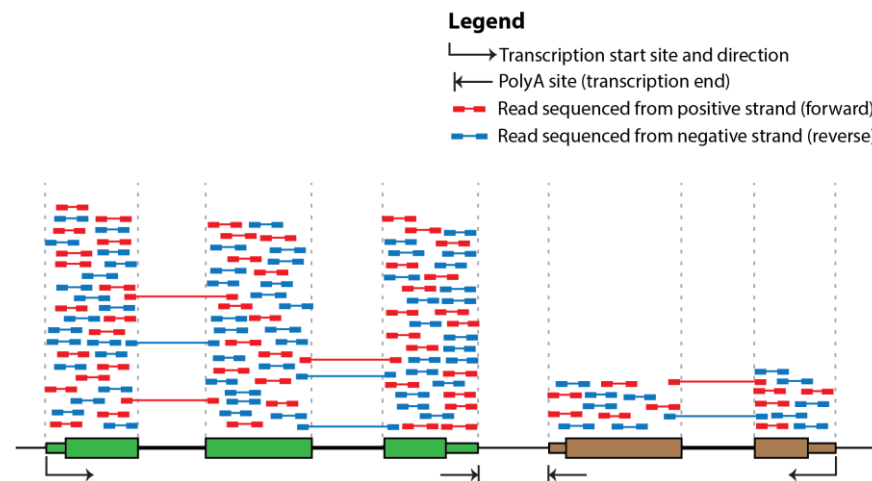
RNA sequence enrichment (selection/depletion)



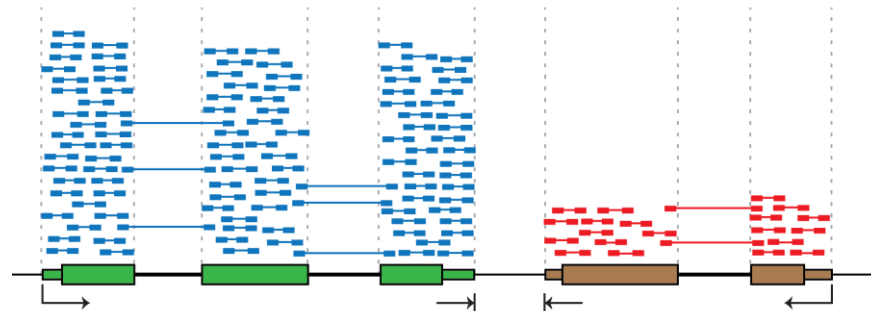


Stranded vs. unstranded

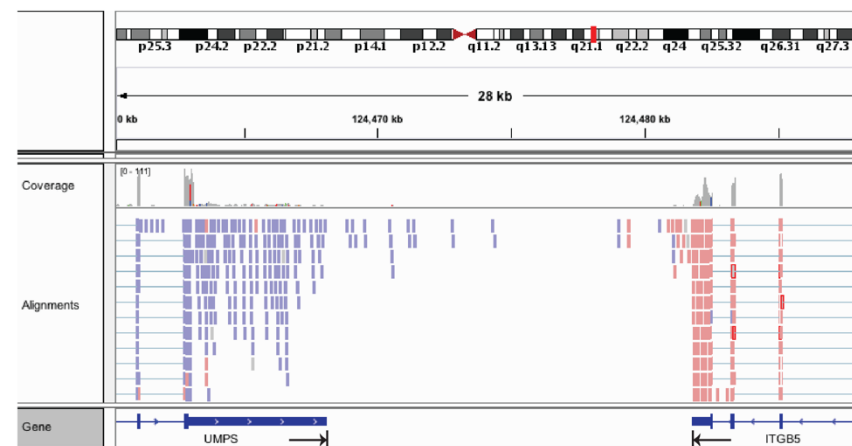
A. Depiction of cDNA fragments from an unstranded library



B. Depiction of cDNA fragments from a stranded library



C. Viewing strand of aligned reads in IGV



<https://rnabio.org/module-09-appendix/0009/12/01/StrandSettings/>
(detailed discussion and cheat sheet)



Ordering RNA-seq data, “coverage”, and cost?

RNA-seq full service, cost per sample ^a	< 12 Samples	≥ 2500 Samples*	~Targeted Coverage
PolyA selection	\$287	\$215	30M reads
Ribosomal depletion, RiboErase (H/M/R)	\$297	\$226	30M reads
Ribosomal depletion, FastSelect (H/M/R)	\$268	\$205	30M reads
Ribosomal depletion, FastSelect (H/M/R+Globin)	\$272	\$213	30M reads
Ribosomal depletion, Watchmaker (H/M/R+Globin)	\$291	NA	30M reads
Low input - Takara SMARTseq mRNA	\$267	\$203	30M reads
Low input - Sigma Seqplex	\$273	\$202	30M reads

- An example menu from a sequencing core facility (circa 2024)
- Options primarily relate to method of enrichment and input amounts
- “Coverage” is a non-intuitive concept in bulk-RNAseq.
 - 30M reads is sufficient for gene abundance estimation (increase for other applications)

Common analysis goals of RNA-Seq analysis (what can you ask of the data?)



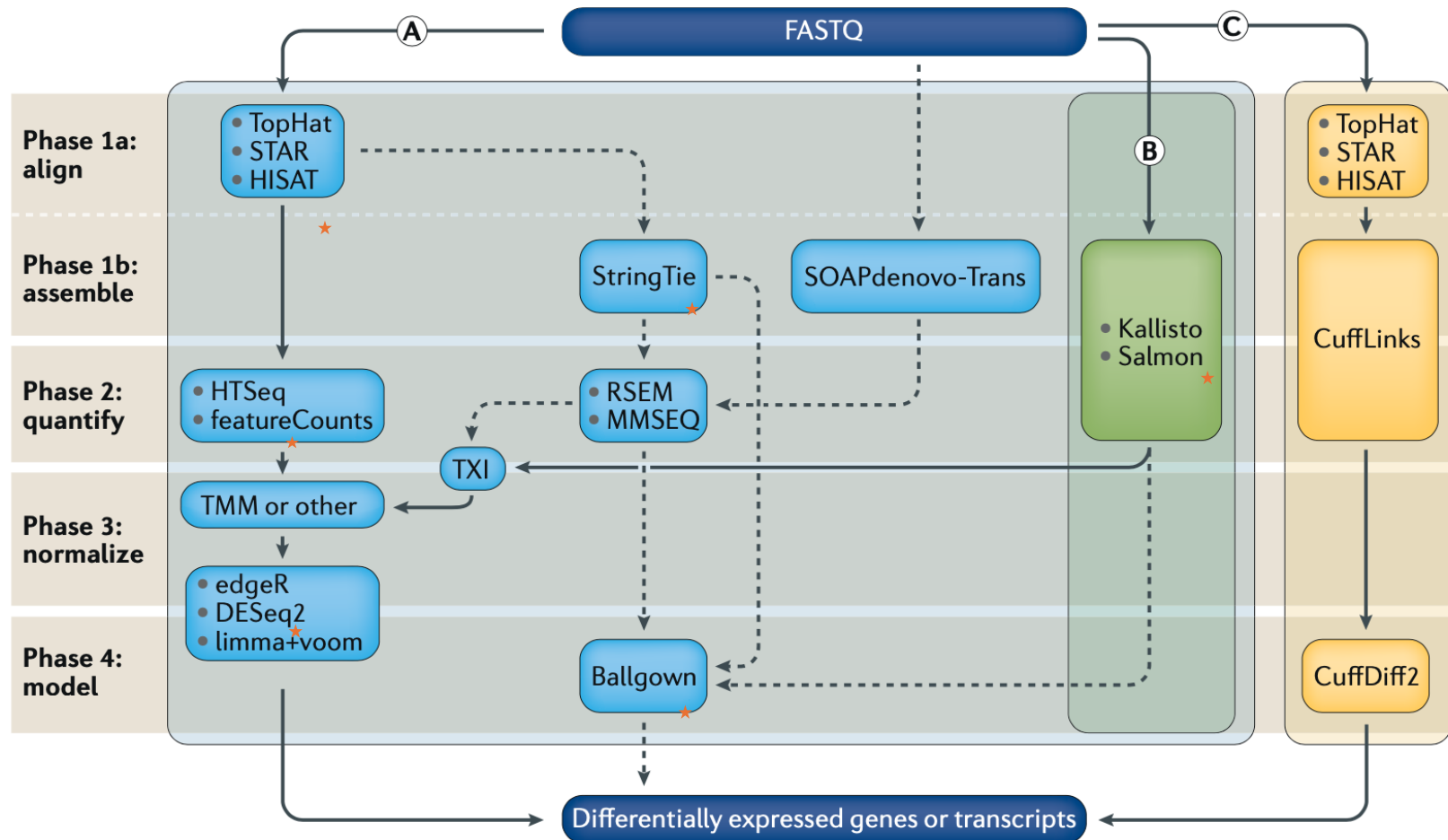
- Gene expression and differential expression
- Alternative expression analysis
- Transcript discovery and annotation
- Allele specific expression
 - Relating to SNPs or mutations
- Mutation discovery
- Fusion detection
- RNA editing

General themes of RNA-seq workflows



- Each type of RNA-seq analysis has distinct requirements and challenges but also a common theme:
 1. Obtain raw data (convert format)
 2. Align/assemble reads
 3. Process alignment with a tool specific to the goal
 - e.g. ‘cufflinks’ for expression analysis, ‘defuse’ for fusion detection, etc.
 4. Post process
 - Import into downstream software (R, Matlab, Cytoscape, Ingenuity, etc.)
 5. Summarize and visualize
 - Create gene lists, prioritize candidates for validation, etc.

Examples of RNA-seq data analysis workflows for differential gene expression



Discussion of bulk vs single cell RNA-seq

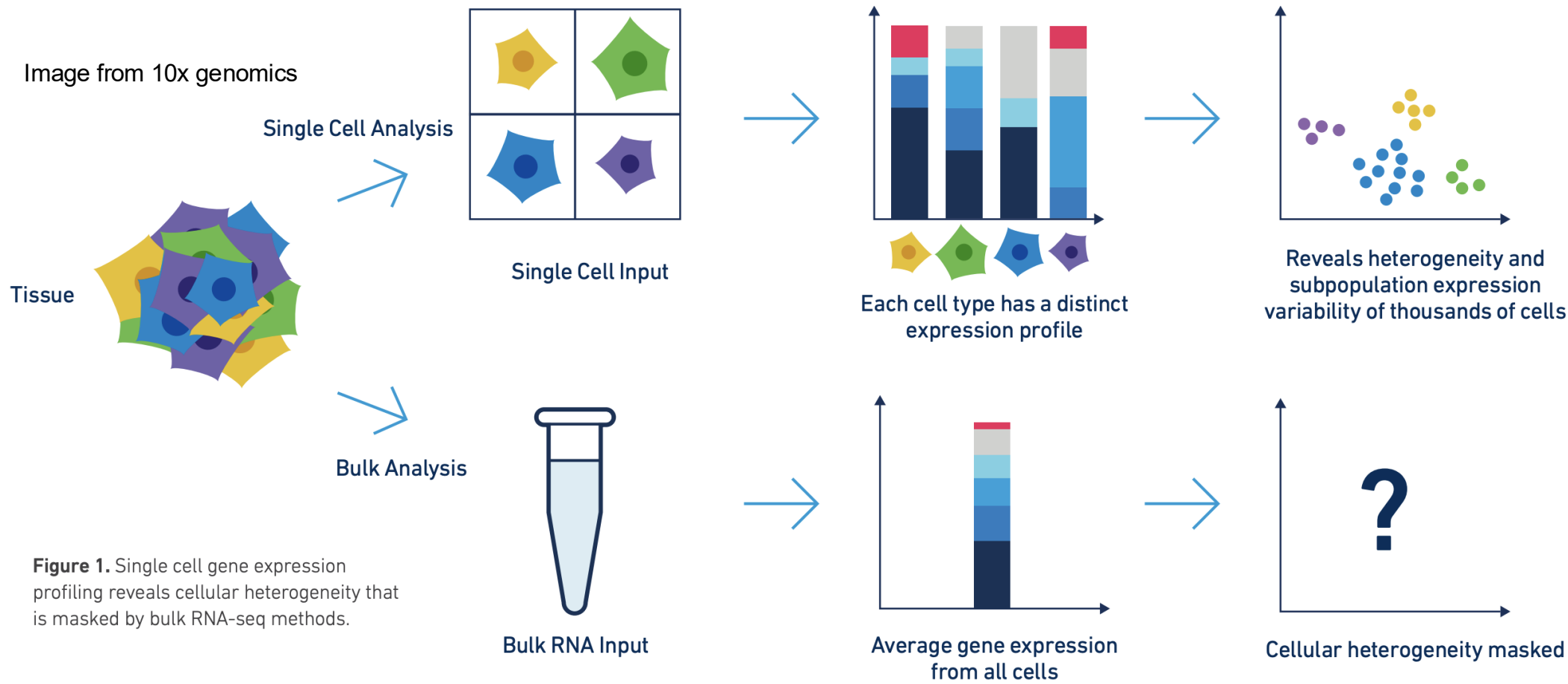


Figure 1. Single cell gene expression profiling reveals cellular heterogeneity that is masked by bulk RNA-seq methods.

Factors to compare: Cost, complexity of library prep, complexity of analysis, qualitative and quantitative differences in richness of information obtained.

Common questions (and answers)

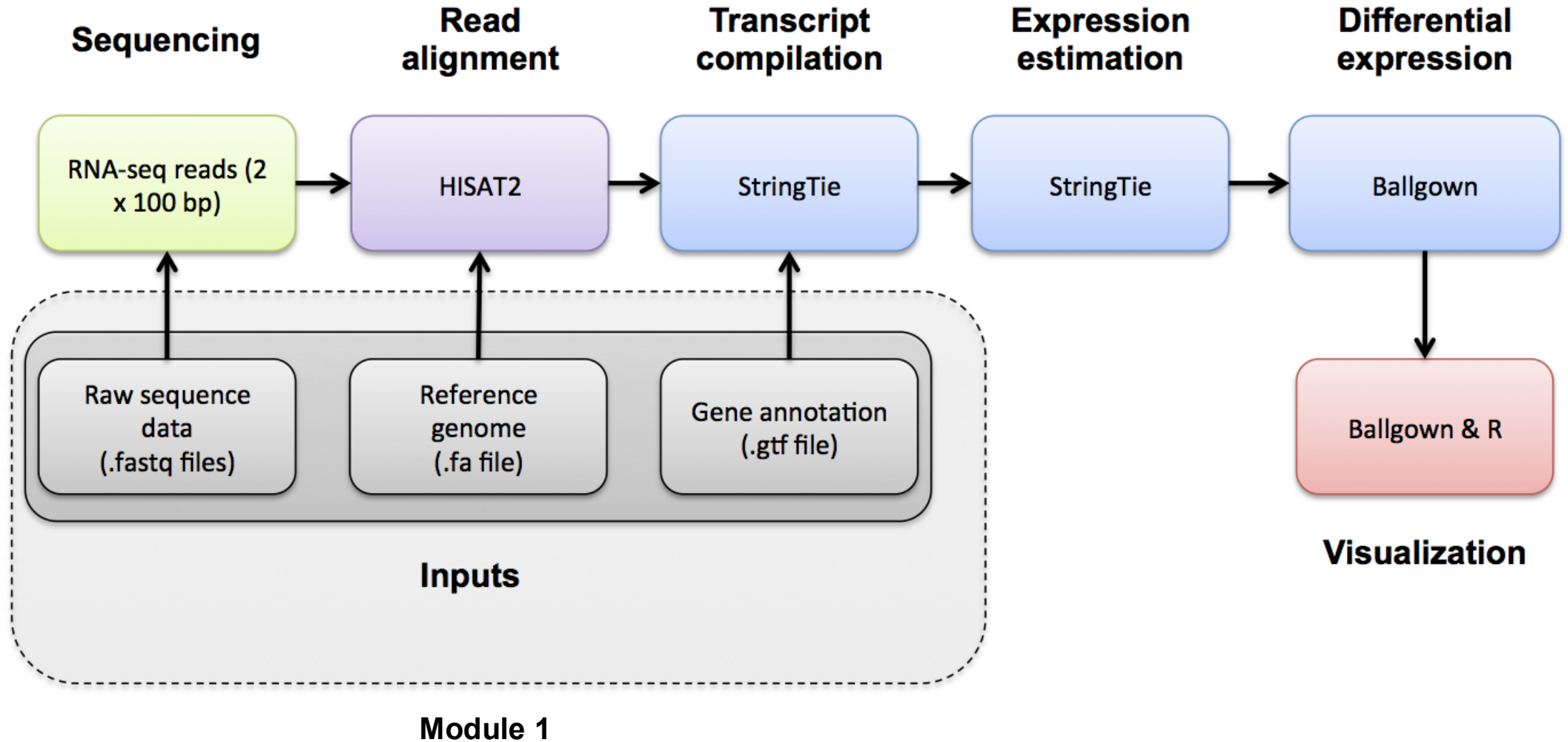


- [Supplementary Table 7](#)
- Malachi Griffith*, Jason R. Walker, Nicholas C. Spies, Benjamin J. Ainscough, Obi L. Griffith*. 2015. Informatics for RNA-seq: A web resource for analysis on the cloud. 11(8):e1004393. 2015.
 - <http://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1004393>



Introduction to tutorial (Module 1)

HISAT2/StringTie/Ballgown RNA-seq Pipeline



Bioinformatics troubleshooting cheat sheet



- ❑ Check your inputs!
- ❑ Mix of incompatible reference genomes used (see [this tutorial](#))
- ❑ Mix of incompatible gene/transcript identifiers
- ❑ Reference sequence names (e.g. “1” vs “chr1”)
- ❑ 1-based vs 0-based coordinates (see [this tutorial](#))
- ❑ Computational tasks fail due to resource limitations (memory and storage)
- ❑ Dependency hell for bioinformatics tools. Learn to use containers (e.g. docker) or environment managers (e.g. conda)



Coffee Break & Networking Session

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